ISP560 PRESENTATION

BPNN Application for Predicting Online Shoppers' Purchasing Intention



Presented By

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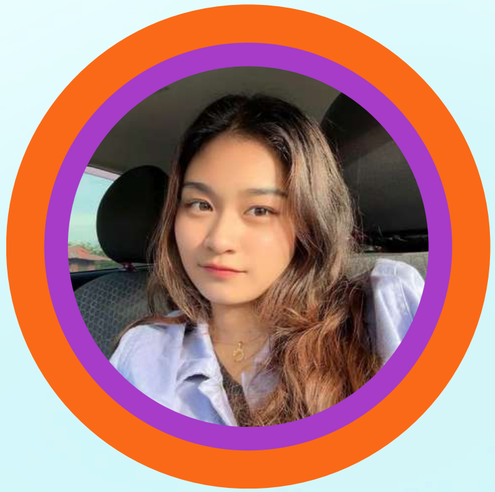
Prepared for: Dr. Khairul Adilah Ahmad

Group Members





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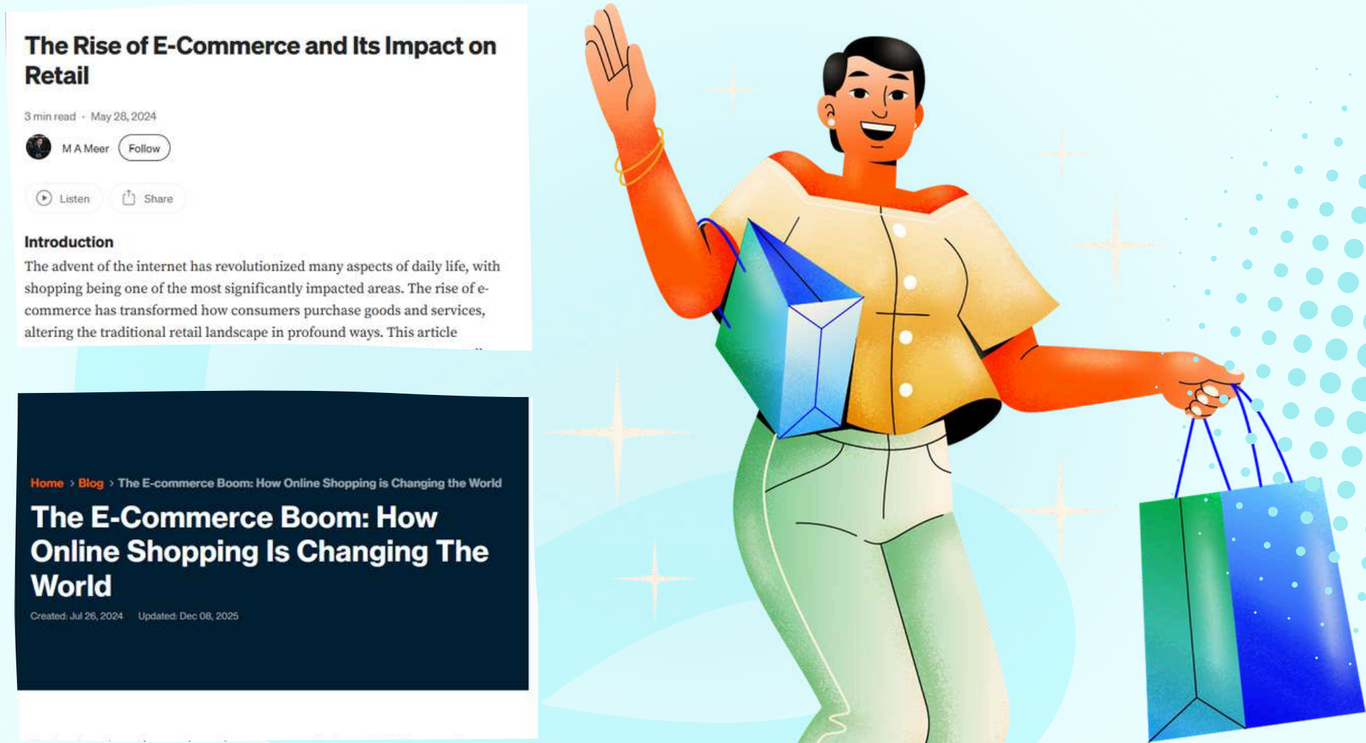
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Research Background

- E-commerce has transformed global retail
- understanding user behavior is a competitive necessity.





Motivation

- The ability to predict a visitor's "purchase intent" in real-time is valuable! .



Why This Dataset?

- everyone wants to know: "Are the customers going to buy it, or are they just wasting my bandwidth?"

"Guessing is expensive but predicting is profitable."

MAIN IDEA

CHALLENGES

The core problem
wasted resources on marketing to buyers with 0-intention of buying

The technical gap
traditional linear models classify "Revenues" outcomes inaccurately

The need
robust, multi-layered NN to predict purchase intent

FACTORS

Data Volume
massive amounts of data; impossible to analyze manually

Non-Linear Patterns
shopping behavior is NOT a straight line

Important Aspects
Temporal Aspect: special days, months, weekend visits

Temporal Aspect: special days, months, weekend visits

SIGNIFICANCE

01 eliminate noisy factors; ensuring the model focuses only on relevant shopper characteristics.

02 allows the network to learn the intricate relationship between time spent on site and actual revenue generation.

03 model provide a reliable analysis that a business can actually use for decision-making.

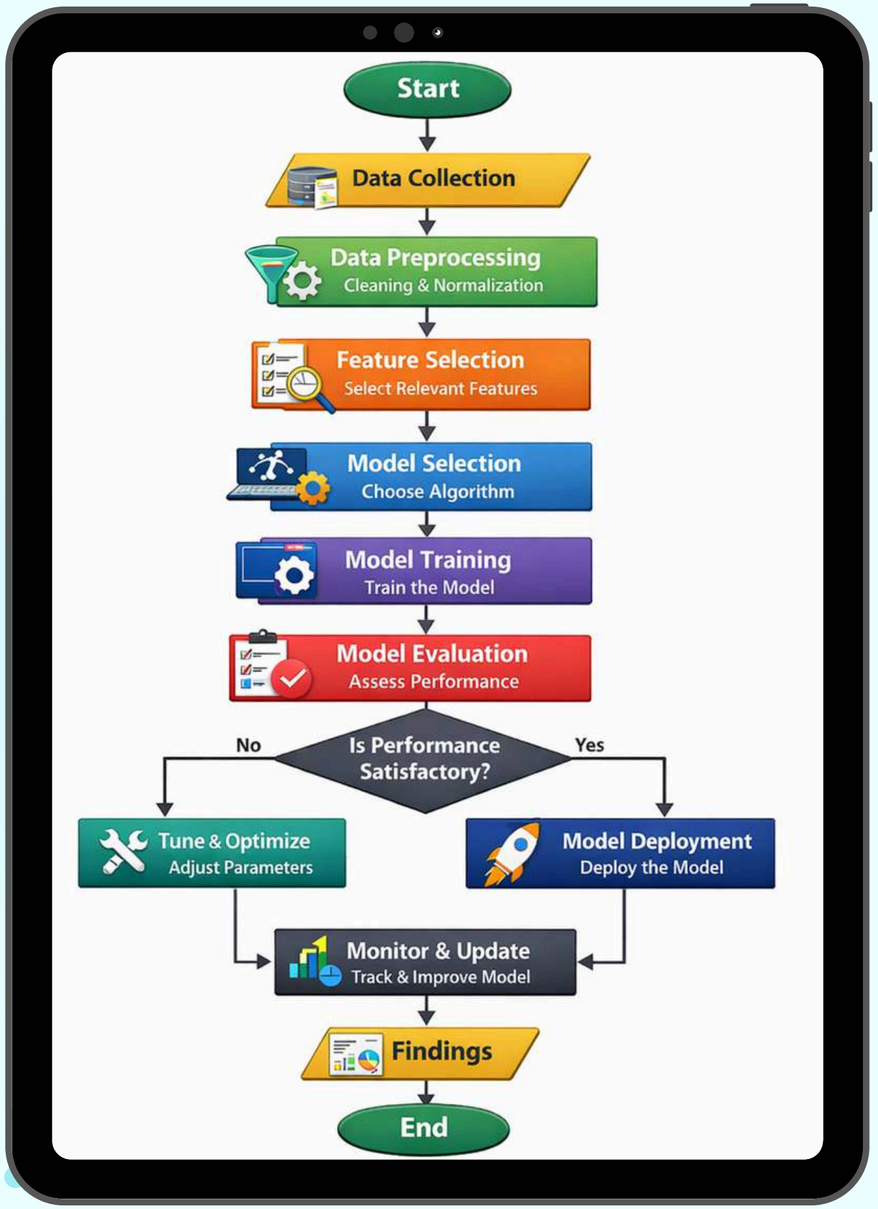
OBJECTIVES

01 To analyze the "Online Shoppers Purchasing Intention" dataset to establish a high-quality data foundation for revenue prediction.

02 To design a multi-layered BPNN featuring three hidden layers, optimized through fine-tuned learning rates and momentum to ensure efficient weight convergence.

03 To evaluate model generalization on a 20% unseen test set using performance metrics, including Accuracy, Kappa Statistic, and F-Measure, to ensure reliable predictive intelligence.

Predictive Modelling Methodology



Frank, E., Hall, M. A., & Witten, I. H. (2016). The WEKA Workbench. Online Appendix for "Data Mining: Practical Machine Learning Tools and Techniques" (4th ed.). Morgan Kaufmann.

Data Identification & Characterization

No.	Attribute Name	Type	Description
1	Administrative	Numerical	Number of administrative pages visited by the visitor
2	Administrative_Duration	Numerical	Total time spent on administrative pages (in seconds)
3	Informational	Numerical	Number of informational pages visited by the visitor
4	Informational_Duration	Numerical	Total time spent on informational pages (in seconds)
5	ProductRelated	Numerical	Number of product-related pages visited by the visitor
6	ProductRelated_Duration	Numerical	Total time spent on product-related pages (in seconds)
7	BounceRates	Numerical	Percentage of visitors who enter the site and leave without triggering other requests
8	ExitRates	Numerical	Percentage of pageviews that were the last in the session
9	PageValues	Numerical	Average value for a page visited before completing a transaction
10	SpecialDay	Numerical	Closeness of site visiting time to special days (0 to 1)
11	Month	Categorical	Month of the year when the session occurred
12	OperatingSystems	Categorical	Operating system of the visitor
13	Browser	Categorical	Browser used by the visitor
14	Region	Categorical	Geographic region of the visitor
15	TrafficType	Categorical	Traffic source that led the visitor to the website
16	Region_Product_Intent_Flag	Categorical	Flag indicating product intent by region
17	Peak_Traffic_Season	Categorical	Indicator of whether session occurred during peak traffic season
18	VisitorType	Categorical	Whether visitor is returning or new (implied from standard attributes)
19	Weekend	Categorical	Boolean indicating if the visit occurred on a weekend
20	Revenue	Categorical	Class label: whether the session ended with a transaction (True/False)

Dataset Information:

- Total Instances: 12,330 sessions
- Positive Class (Revenue=True): 1,908 sessions (15.5%)
- Negative Class (Revenue=False): 10,422 sessions (84.5%)
- Time Period: 1 year
- Missing Values: None

Full Dataset Loaded in WEKA

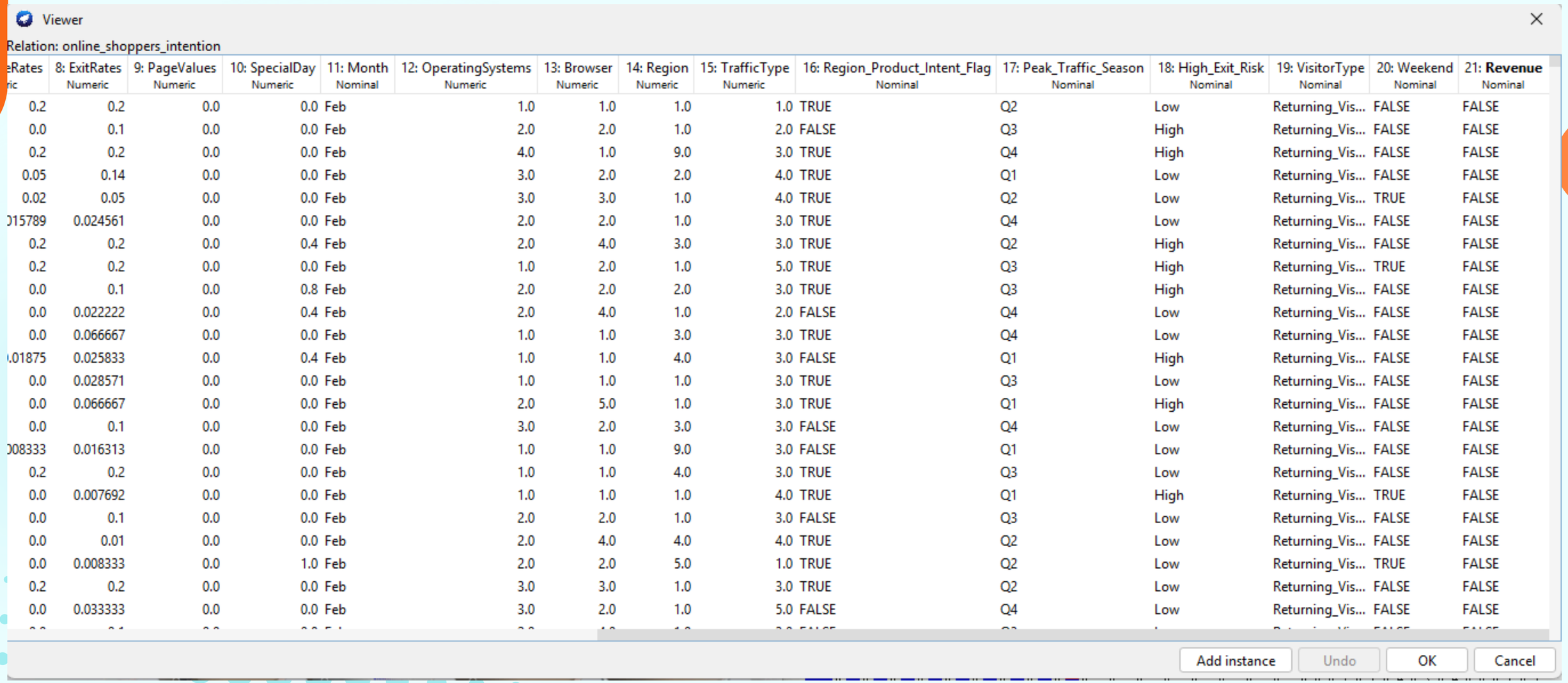
Viewer

Relation: online_shoppers_intention

No.	1: Administrative Numeric	2: Administrative_Duration Numeric	3: Informational Numeric	4: Informational_Duration Numeric	5: ProductRelated Numeric	6: ProductRelated_Duration Numeric	7: BounceRates Numeric	8: ExitRates Numeric	9: PageValues Numeric	10: SpecialDay Numeric	11: Month Nominal	12: OperatingSystems Numeric	13: Brow Numeric
1	0.0	0.0	0.0	0.0	1.0	0.0	0.2	0.2	0.0	0.0	Feb	1.0	
2	0.0	0.0	0.0	0.0	2.0	64.0	0.0	0.1	0.0	0.0	Feb	2.0	
3	0.0	0.0	0.0	0.0	1.0	0.0	0.2	0.2	0.0	0.0	Feb	4.0	
4	0.0	0.0	0.0	0.0	2.0	2.666667	0.05	0.14	0.0	0.0	Feb	3.0	
5	0.0	0.0	0.0	0.0	10.0	627.5	0.02	0.05	0.0	0.0	Feb	3.0	
6	0.0	0.0	0.0	0.0	19.0	154.216667	0.015789	0.024561	0.0	0.0	Feb	2.0	
7	0.0	0.0	0.0	0.0	1.0	0.0	0.2	0.2	0.0	0.4	Feb	2.0	
8	1.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	Feb	1.0	
9	0.0	0.0	0.0	0.0	2.0	37.0	0.0	0.1	0.0	0.8	Feb	2.0	
10	0.0	0.0	0.0	0.0	3.0	738.0	0.0	0.022222	0.0	0.4	Feb	2.0	
11	0.0	0.0	0.0	0.0	3.0	395.0	0.0	0.066667	0.0	0.0	Feb	1.0	
12	0.0	0.0	0.0	0.0	16.0	407.75	0.01875	0.025833	0.0	0.4	Feb	1.0	
13	0.0	0.0	0.0	0.0	7.0	280.5	0.0	0.028571	0.0	0.0	Feb	1.0	
14	0.0	0.0	0.0	0.0	6.0	98.0	0.0	0.066667	0.0	0.0	Feb	2.0	
15	0.0	0.0	0.0	0.0	2.0	68.0	0.0	0.1	0.0	0.0	Feb	3.0	
16	2.0	53.0	0.0	0.0	23.0	1668.285119	0.008333	0.016313	0.0	0.0	Feb	1.0	
17	0.0	0.0	0.0	0.0	1.0	0.0	0.2	0.2	0.0	0.0	Feb	1.0	
18	0.0	0.0	0.0	0.0	13.0	334.966667	0.0	0.007692	0.0	0.0	Feb	1.0	
19	0.0	0.0	0.0	0.0	2.0	32.0	0.0	0.1	0.0	0.0	Feb	2.0	
20	0.0	0.0	0.0	0.0	20.0	2981.166667	0.0	0.01	0.0	0.0	Feb	2.0	
21	0.0	0.0	0.0	0.0	8.0	136.166667	0.0	0.008333	0.0	1.0	Feb	2.0	
22	0.0	0.0	0.0	0.0	2.0	0.0	0.2	0.2	0.0	0.0	Feb	3.0	
23	0.0	0.0	0.0	0.0	3.0	105.0	0.0	0.033333	0.0	0.0	Feb	3.0	

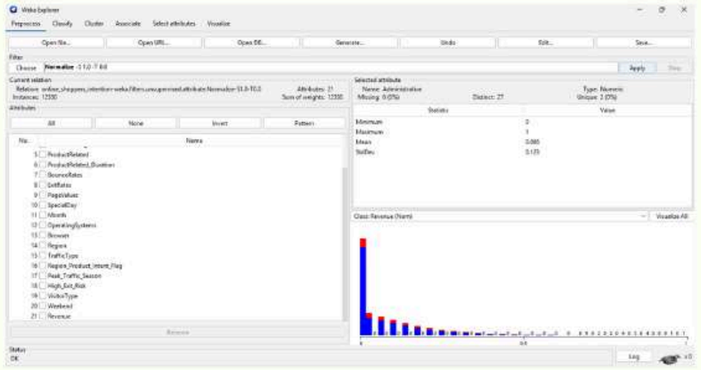
Add instance | Undo | OK | Cancel

Full Dataset Loaded in WEKA



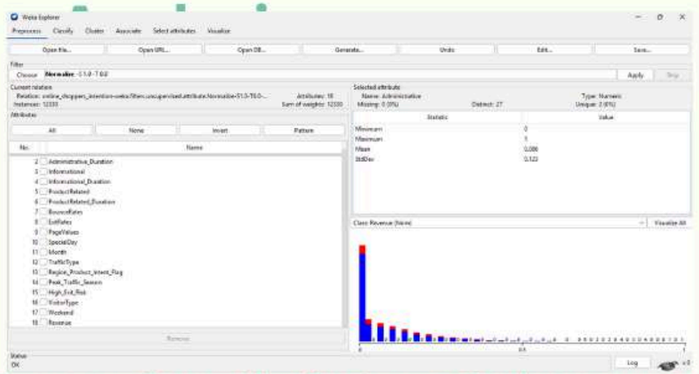
Data Pre-processing & Noise Reduction

Normalization



scaled using the Normalize filter to ensure weight convergence, as recommended by the WEKA documentation (Frank et al., 2016).

Feature Selection: Conducted Information Gain



Browser, OperatingSystems, Region removed

Final Input Count

17 high-relevance attributes remained for the model.

Objective 1: Fullfilled

Data Partitioning

Methodology:

Used the RemovePercentage filter to split the dataset.

Training Set

9,864 instances (80%) used to adjust the network's weights.

Testing Set

2,466 instances (20%) held back for the final evaluation to prove generalization (unseen data).

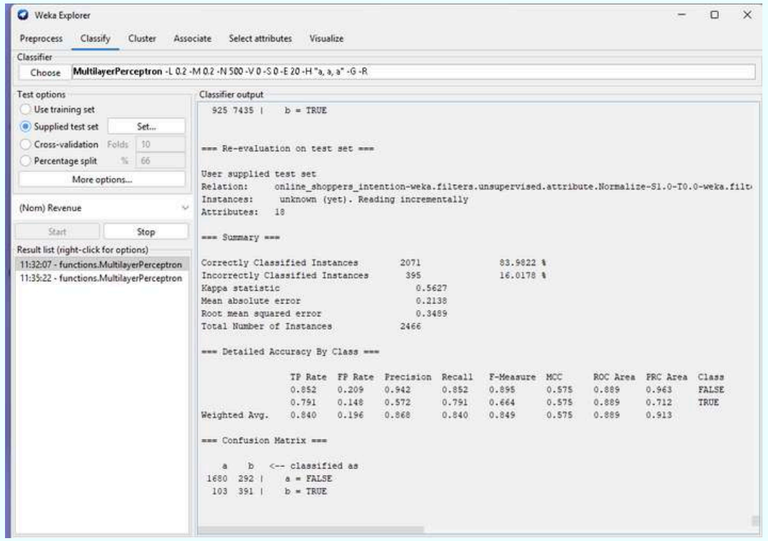


Addressing the Class Imbalance

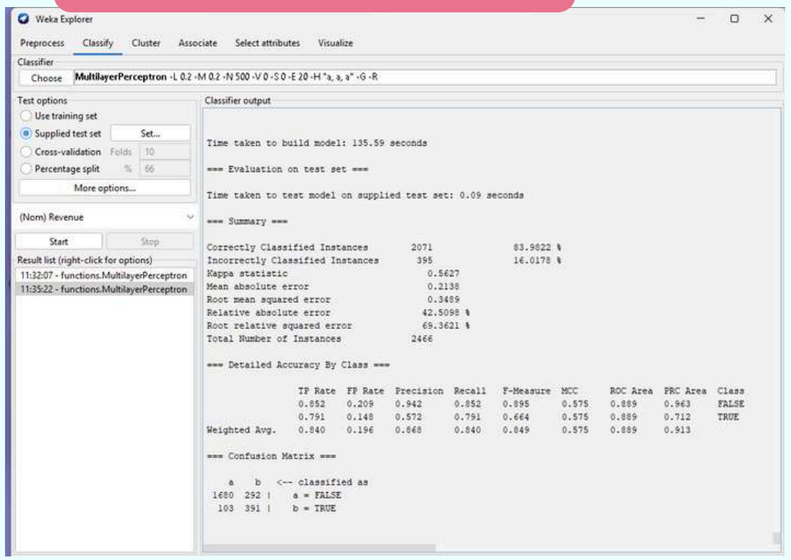
Imbalance Discovery

- 84.5% class imbalance
- to fix: SMOTE filter
- after SMOTE:
- Kappa Statistic (0.5956) and F-Measure (0.66); model's actual predictive intelligence.

Train Set (Balanced)



Test Set (Balanced)

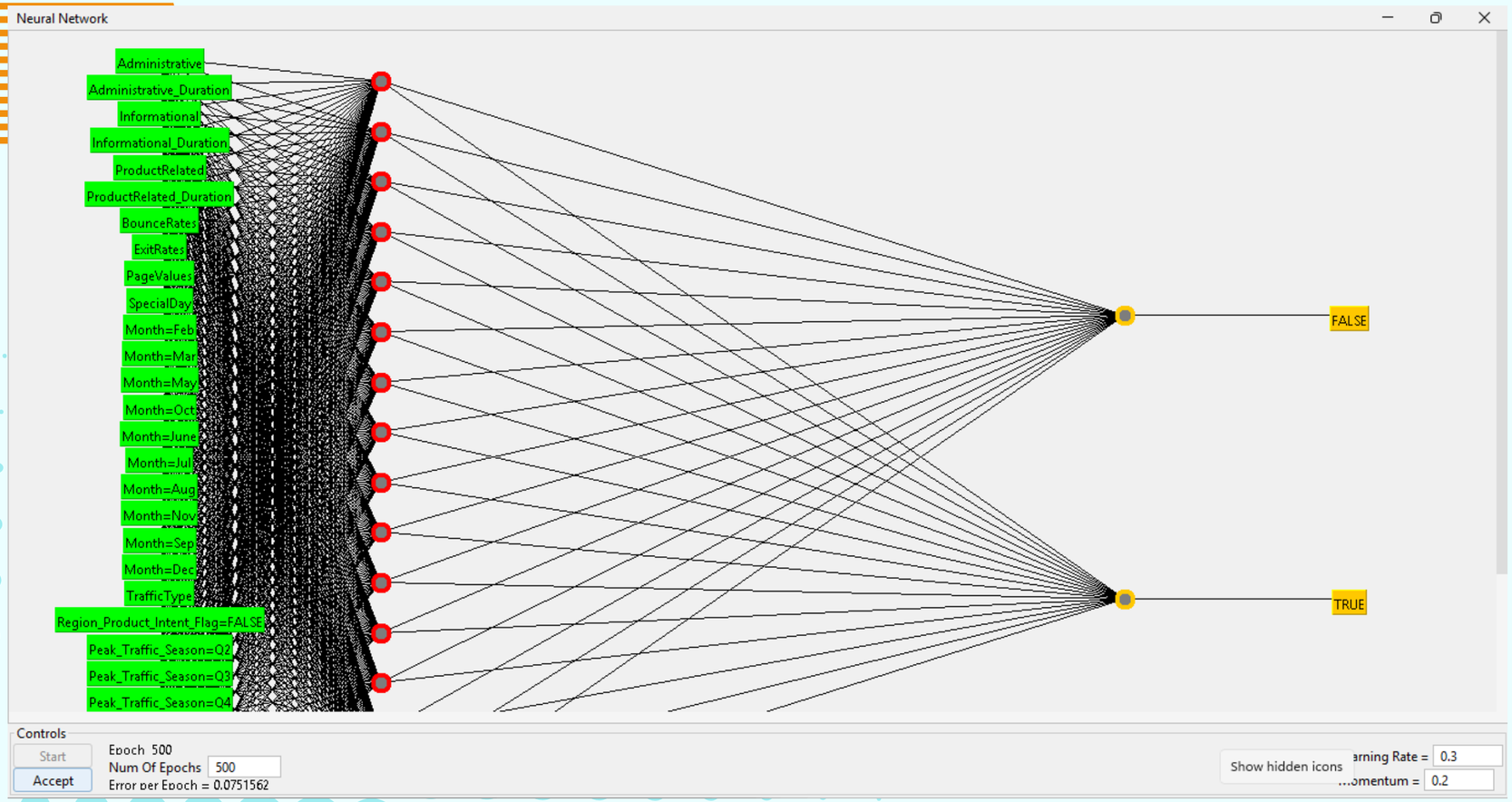


System Functionality - 5 Experiments

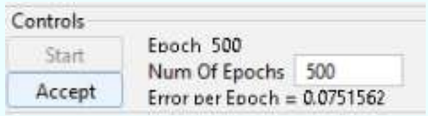


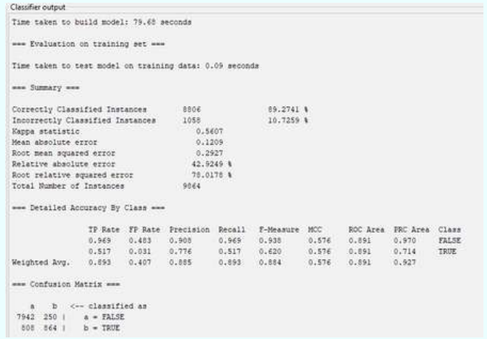
Experiment	Configuration	TRAINING			TESTING		
		Logic	Test Acc	Kappa	Logic	Test Acc	Kappa
1	1 Hidden Layer	Baseline	89.27%	0.56	Baseline	86.05%	0.5
2	3 Hidden Layers	Depth Test	90.76%	0.64	Depth Test	88.60%	0.6
3	LR: 0.2 + 3 Layers	Best Convergence	91.14%	0.65	Best Convergence	88.61%	0.6
4	High Momentum (0.5)	Acceleration	90.98%	0.64	Acceleration	88.28%	0.58
5	Decay: TRUE	Adaptive Rate	89.03%	0.57	Adaptive Rate	86.09%	0.51

Experiment 1 : 1 HL, LR=0.3, M=0.2



Training Set





```
Classifier output

Time taken to build model: 79.68 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.09 seconds

=== Summary ===

Correctly Classified Instances        8806               89.2741 %
Incorrectly Classified Instances      1058               10.7259 %
Kappa statistic                          0.5607
Mean absolute error                      0.1209
Root mean squared error                  0.2927
Relative absolute error                 42.9249 %
Root relative squared error             78.0178 %
Total Number of Instances             9864

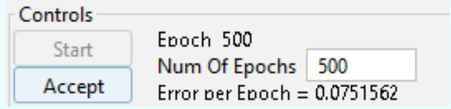
=== Detailed Accuracy By Class ===

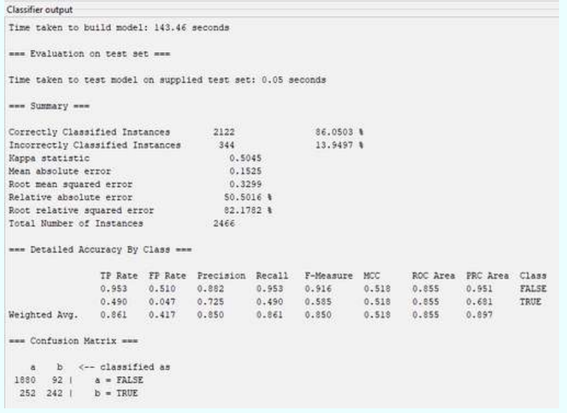
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.969    0.483    0.908      0.969     0.938      0.576     0.891     0.970     FALSE
                 0.517    0.031    0.776      0.517     0.620      0.576     0.891     0.714     TRUE
Weighted Avg.    0.893    0.407    0.885      0.893     0.884      0.576     0.891     0.927

=== Confusion Matrix ===

    a    b   <-- classified as
 7942  250 |    a = FALSE
  808  864 |    b = TRUE
```

Test Set





```
Classifier output

Time taken to build model: 143.46 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.05 seconds

=== Summary ===

Correctly Classified Instances        2122               86.0503 %
Incorrectly Classified Instances       344               13.9497 %
Kappa statistic                          0.5045
Mean absolute error                      0.1525
Root mean squared error                  0.3299
Relative absolute error                 50.5016 %
Root relative squared error             82.1782 %
Total Number of Instances             2466

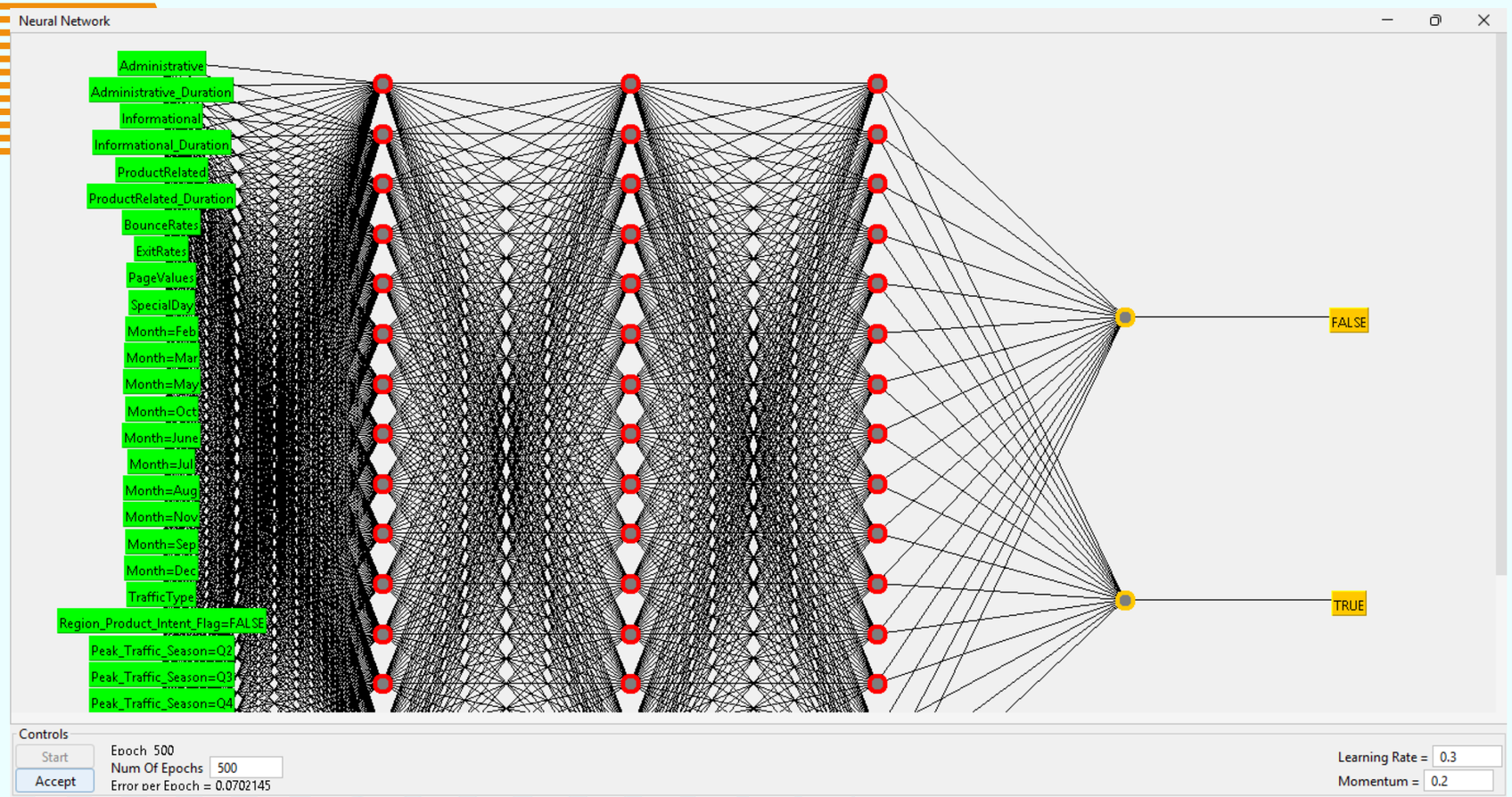
=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.953    0.510    0.882      0.953     0.916      0.518     0.855     0.951     FALSE
                 0.490    0.047    0.725      0.490     0.585      0.518     0.855     0.681     TRUE
Weighted Avg.    0.861    0.417    0.850      0.861     0.850      0.518     0.855     0.897

=== Confusion Matrix ===

    a    b   <-- classified as
 1880   92 |    a = FALSE
  252  242 |    b = TRUE
```

Experiment 2 : 3 HL, LR=0.3, M=0.2



Training Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0702145

Classifier output

```
Time taken to build model: 71.73 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.13 seconds

=== Summary ===

Correctly Classified Instances        8944               90.6732 %
Incorrectly Classified Instances       920                9.3268 %
Kappa statistic                          0.6418
Mean absolute error                      0.1286
Root mean squared error                  0.2701
Relative absolute error                 45.6602 %
Root relative squared error             71.9885 %
Total Number of Instances             9864

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.963    0.370    0.927      0.963    0.945      0.647    0.909     0.974     FALSE
                 0.630    0.037    0.777      0.630    0.696      0.647    0.909     0.759     TRUE
Weighted Avg.    0.907    0.313    0.902      0.907    0.903      0.647    0.909     0.937

=== Confusion Matrix ===

    a    b   <-- classified as
 7890  302 |    a = FALSE
  618 1054 |    b = TRUE
```

Test Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0702145

Classifier output

```
Time taken to build model: 116.42 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.08 seconds

=== Summary ===

Correctly Classified Instances        2185               88.605  %
Incorrectly Classified Instances       281               11.395  %
Kappa statistic                          0.5956
Mean absolute error                      0.1588
Root mean squared error                  0.3056
Relative absolute error                 52.5768 %
Root relative squared error             76.1173 %
Total Number of Instances             2466

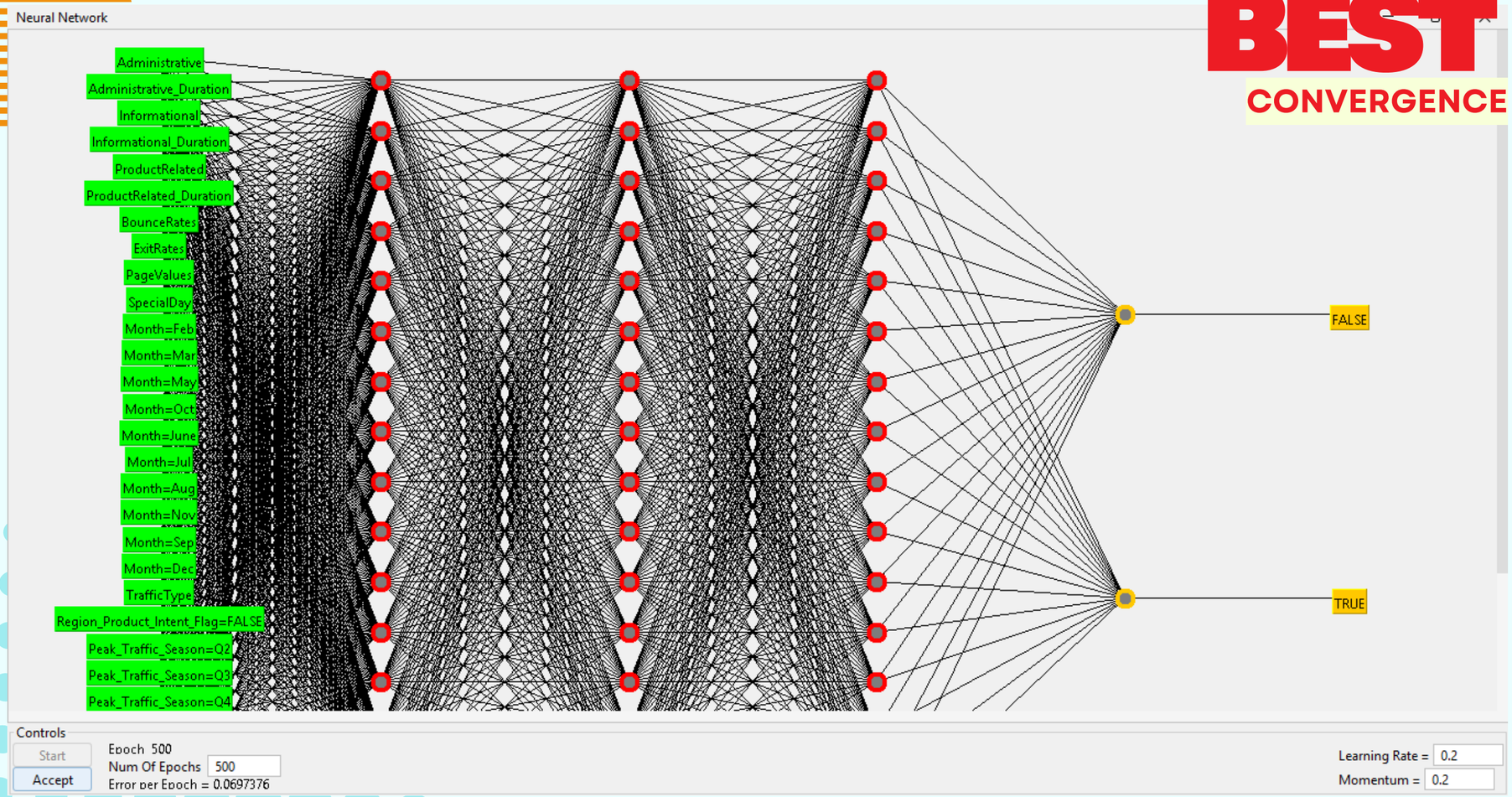
=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.969    0.445    0.897      0.969    0.932      0.612    0.874     0.953     FALSE
                 0.555    0.031    0.818      0.555    0.661      0.612    0.874     0.727     TRUE
Weighted Avg.    0.886    0.362    0.881      0.886    0.877      0.612    0.874     0.908

=== Confusion Matrix ===

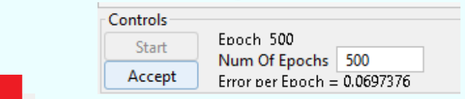
    a    b   <-- classified as
 1911   61 |    a = FALSE
  220  274 |    b = TRUE
```

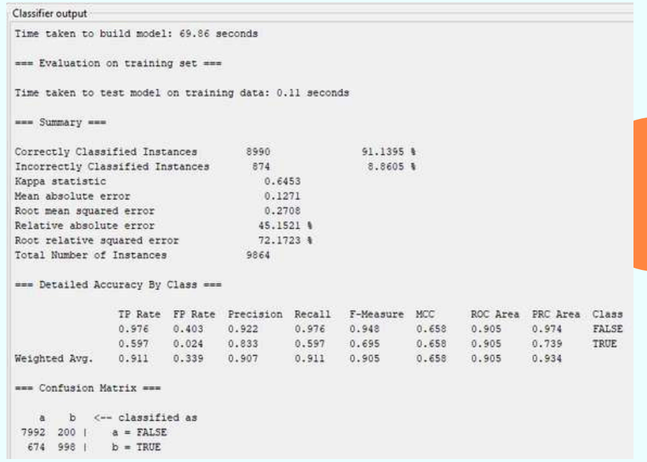
Experiment 3 : 3 HL, LR=0.2, M=0.2



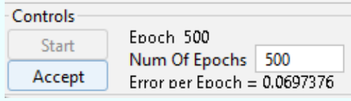
BEST CONVERGENCE

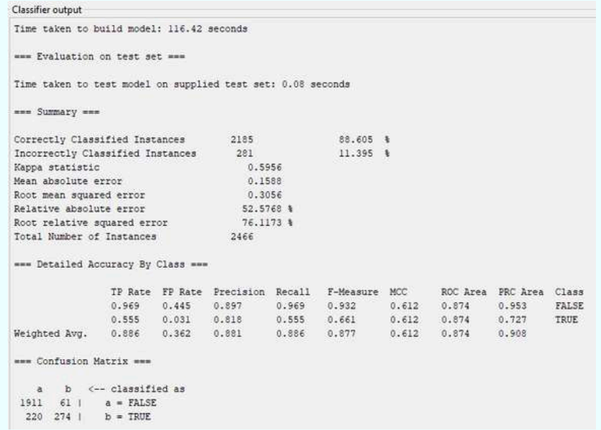
Training Set





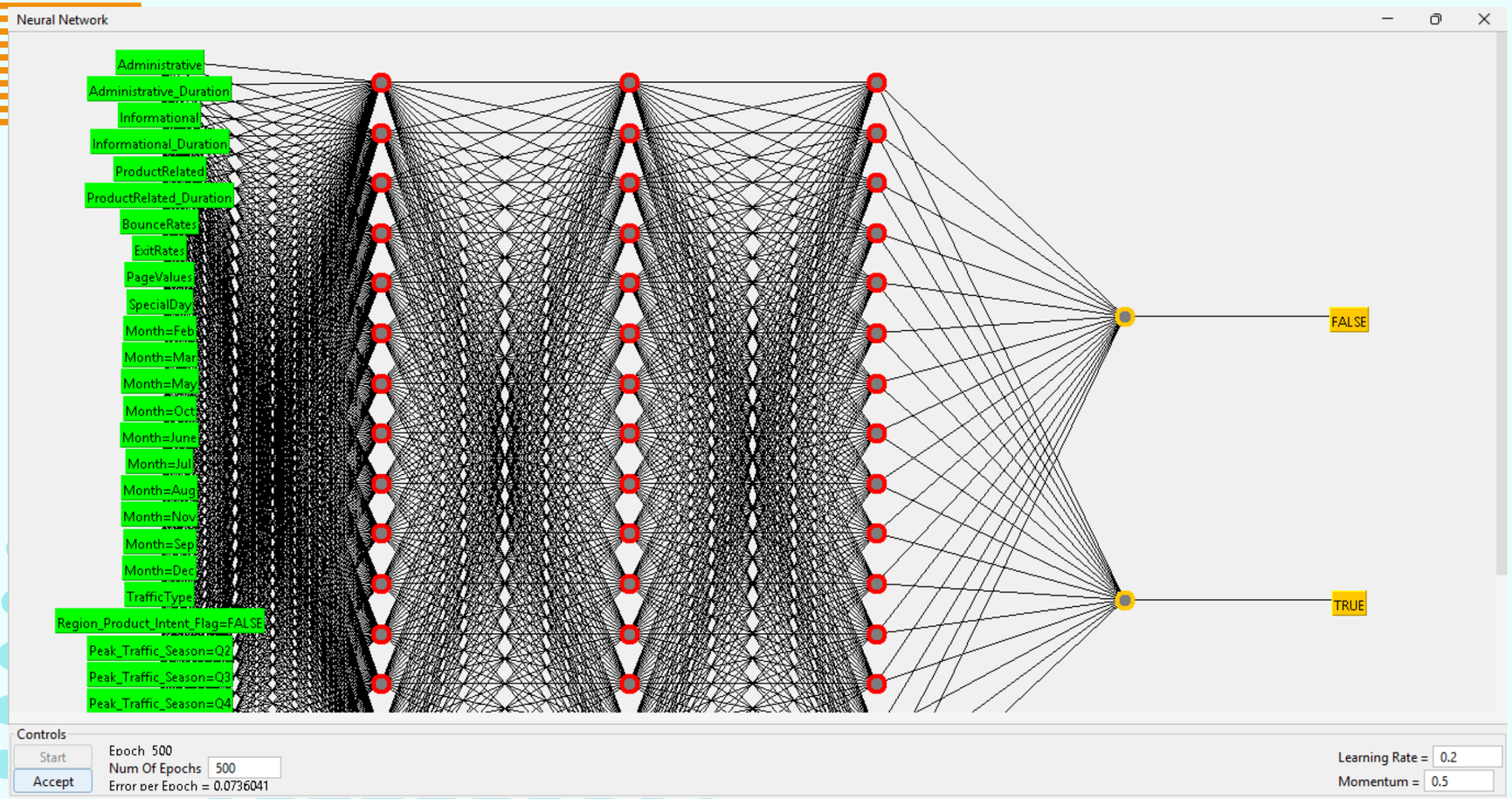
Test Set





Objective 2: Fullfilled

Experiment 4 : 3 HL, LR=0.2, M=0.5



Training Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0736041

Classifier output

```
Time taken to build model: 75.03 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.13 seconds

=== Summary ===

Correctly Classified Instances        8974               90.9773 %
Incorrectly Classified Instances       890                9.0227 %
Kappa statistic                          0.6374
Mean absolute error                      0.129
Root mean squared error                  0.2705
Relative absolute error                 45.8144 %
Root relative squared error             72.0856 %
Total Number of Instances             9864

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.975    0.412    0.921     0.975     0.947      0.650    0.903     0.973     FALSE
                 0.588    0.025    0.830     0.588     0.688      0.650    0.903     0.747     TRUE
Weighted Avg.    0.910    0.346    0.905     0.910     0.903      0.650    0.903     0.935

=== Confusion Matrix ===

    a    b   <-- classified as
 7991  201 |    a = FALSE
  689  983 |    b = TRUE
```

Test Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0697376

Classifier output

```
Time taken to build model: 176.06 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.16 seconds

=== Summary ===

Correctly Classified Instances        2177               88.2806 %
Incorrectly Classified Instances       289               11.7194 %
Kappa statistic                          0.5752
Mean absolute error                      0.1568
Root mean squared error                  0.3043
Relative absolute error                 51.9209 %
Root relative squared error             75.8107 %
Total Number of Instances             2466

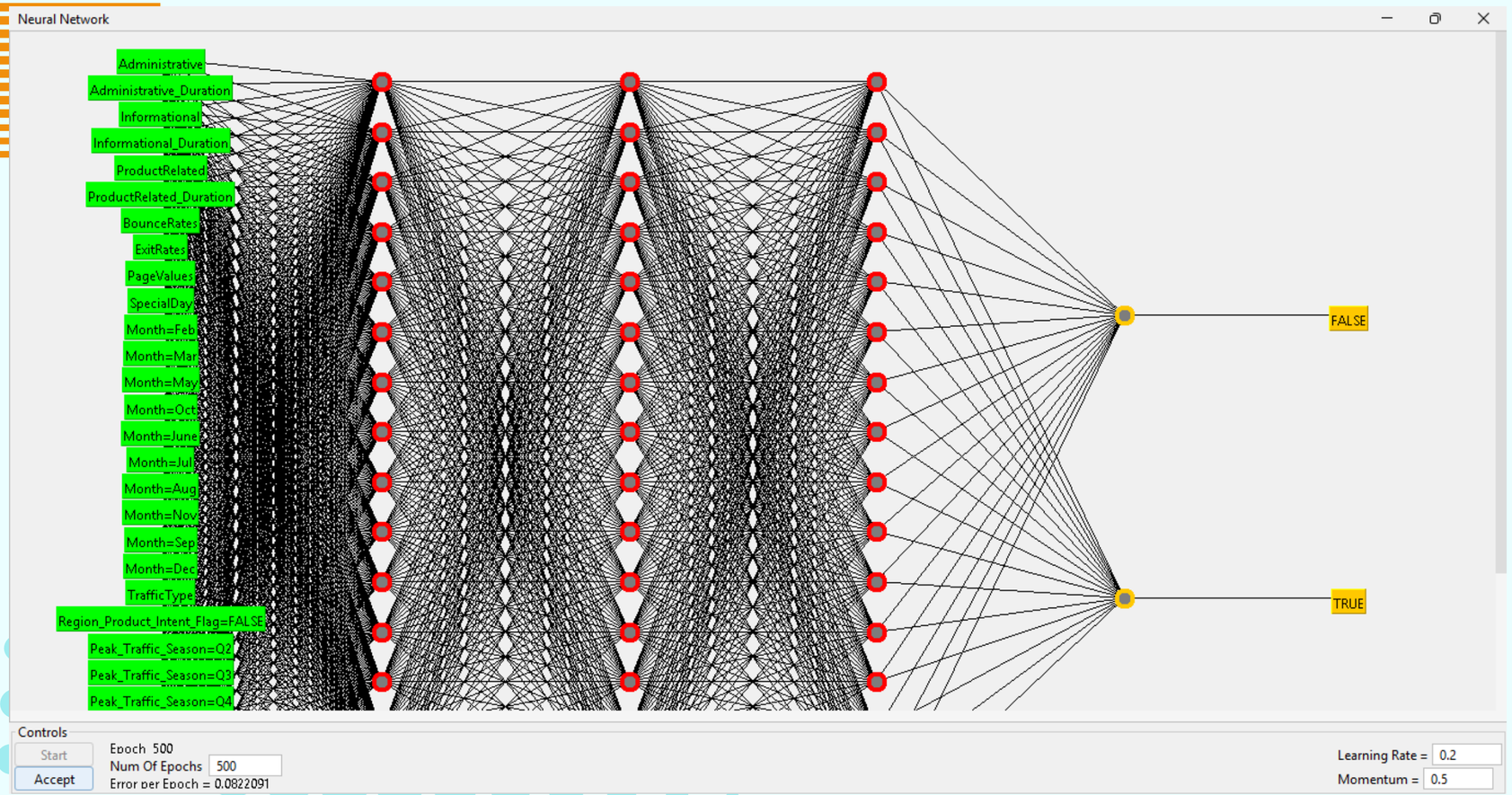
=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.973    0.478    0.890     0.973     0.930      0.597    0.877     0.957     FALSE
                 0.522    0.027    0.830     0.522     0.641      0.597    0.877     0.748     TRUE
Weighted Avg.    0.883    0.387    0.878     0.883     0.872      0.597    0.877     0.915

=== Confusion Matrix ===

    a    b   <-- classified as
 1919   53 |    a = FALSE
  236  258 |    b = TRUE
```

Experiment 5 : 3 HL, LR=0.2, M=0.5, Decay: TRUE



Training Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0822091

```
=== Evaluation on training set ===

Time taken to test model on training data: 0.31 seconds

=== Summary ===

Correctly Classified Instances        8782               89.0308 %
Incorrectly Classified Instances      1082               10.9692 %
Kappa statistic                          0.5696
Mean absolute error                      0.1665
Root mean squared error                  0.2862
Relative absolute error                 59.1363 %
Root relative squared error             76.2829 %
Total Number of Instances             9864

=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.958    0.443    0.914      0.958   0.936      0.577    0.899     0.973     FALSE
               0.557    0.042    0.732      0.557   0.633      0.577    0.899     0.690     TRUE
Weighted Avg.  0.890    0.375    0.883      0.890   0.884      0.577    0.899     0.925

=== Confusion Matrix ===

    a    b   <-- classified as
 7850  342 |    a = FALSE
  740  932 |    b = TRUE
```

Test Set

Controls

Start

Accept

Epoch 500

Num Of Epochs 500

Error per Epoch = 0.0822091

Classifier output

```
Time taken to build model: 138.04 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.13 seconds

=== Summary ===

Correctly Classified Instances        2123               86.0908 %
Incorrectly Classified Instances       343               13.9092 %
Kappa statistic                          0.5089
Mean absolute error                      0.2018
Root mean squared error                  0.3167
Relative absolute error                 66.844  %
Root relative squared error             78.8866 %
Total Number of Instances             2466

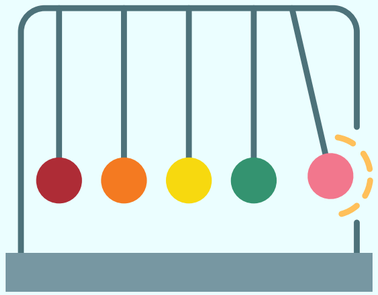
=== Detailed Accuracy By Class ===

               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.952    0.502    0.883      0.952   0.916      0.522    0.875     0.957     FALSE
               0.498    0.048    0.721      0.498   0.589      0.522    0.875     0.680     TRUE
Weighted Avg.  0.861    0.411    0.851      0.861   0.851      0.522    0.875     0.902

=== Confusion Matrix ===

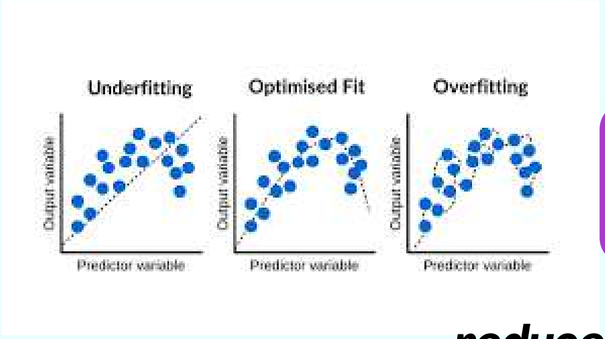
    a    b   <-- classified as
 1877   95 |    a = FALSE
  248  246 |    b = TRUE
```

Applied Enhancement: BPNN with Momentum and Adaptive Learning Rate (Decay).



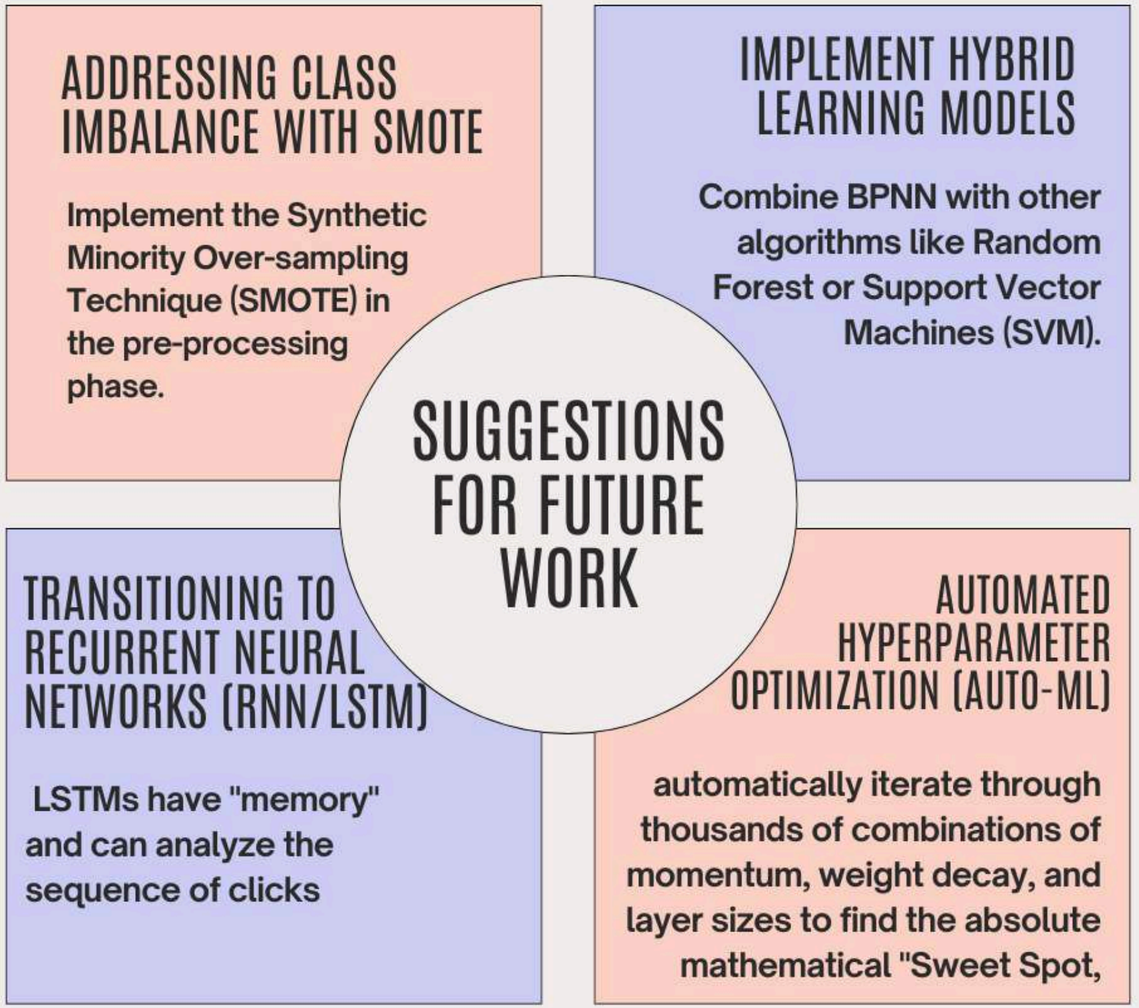
Momentum = 0.2

to help the network "push" through local minima in the error surface.



Decay

reduce the learning rate over time, allowing the model to fine-tune weights as it reached the global minimum.

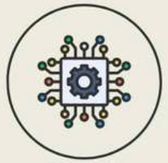

SUGGESTIONS FOR FUTURE WORK
ADDRESSING CLASS IMBALANCE WITH SMOTE
Implement the Synthetic Minority Over-sampling Technique (SMOTE) in the pre-processing phase.
IMPLEMENT HYBRID LEARNING MODELS
Combine BPNN with other algorithms like Random Forest or Support Vector Machines (SVM).
TRANSITIONING TO RECURRENT NEURAL NETWORKS (RNN/LSTM)
LSTMs have "memory" and can analyze the sequence of clicks
AUTOMATED HYPERPARAMETER OPTIMIZATION (AUTO-ML)
automatically iterate through thousands of combinations of momentum, weight decay, and layer sizes to find the absolute mathematical "Sweet Spot,

Project's Findings

Focuses on the rigorous analysis and evaluation of the Backpropagation Neural Network's (BPNN) predictive performance

At the end of this project we found:

1. OVERALL CLASSIFICATION ACCURACY



High Predictive Success: The final BPNN model achieved an overall testing accuracy of 88.61% on the unseen 20% data partition

2. STATISTICAL RELIABILITY (KAPPA STATISTIC)



Beyond Random Chance: The model achieved a Kappa Statistic of 0.5956.

3. CLASS-SPECIFIC PERFORMANCE (TRUE VS. FALSE)



the performance metrics for the TRUE (Buyer) class are the most critical findings; a precision of 0.833 for the TRUE class, meaning that when the model predicts a user will buy, it is correct 83.3% of the time.

4. ERROR ANALYSIS AND CONVERGENCE



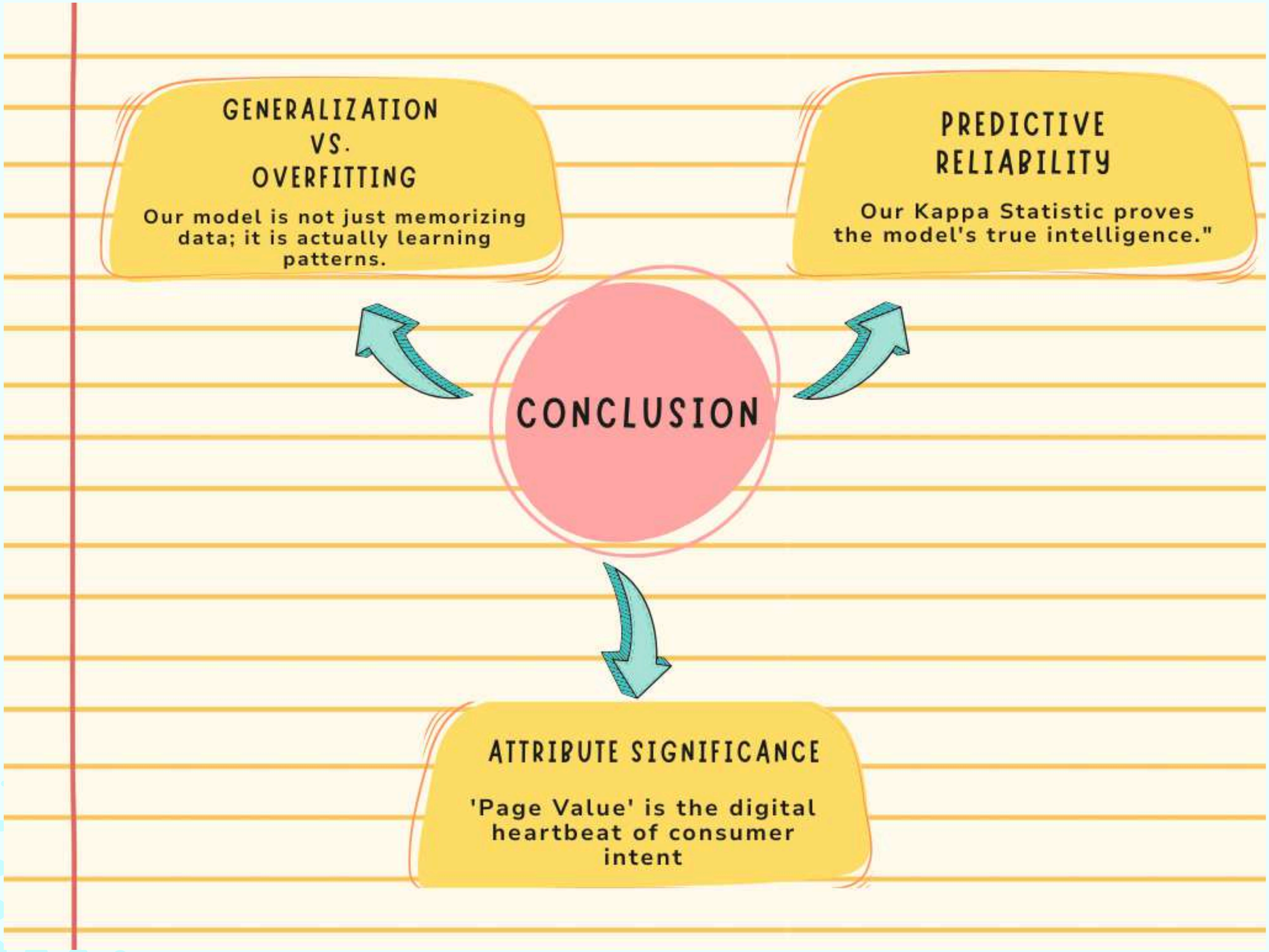
Mean Absolute Error (MAE): The model recorded a low MAE of 0.2133, indicating that the average magnitude of the errors in the predictions was small.

5. IDENTIFYING THE "DIGITAL HEARTBEAT"



Attribute Significance: A key qualitative finding was that 'Page Value' served as the primary indicator for revenue generation.

Objective 3: Fullfilled


GENERALIZATION
VS.
OVERFITTING
Our model is not just memorizing data; it is actually learning patterns.
PREDICTIVE
RELIABILITY
Our Kappa Statistic proves the model's true intelligence."
CONCLUSION
ATTRIBUTE SIGNIFICANCE
'Page Value' is the digital heartbeat of consumer intent

References

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- Ahmad, A. (2025). ISP560: Machine Learning - Chapter 2: Supervised Learning (Backpropagation) [Lecture Notes]. College of Computing, Informatics and Mathematics, Universiti Teknologi MARA.
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THANK YOU

so much for your attention :)

